EVALUATING THE IMPACT OF DEDICATED BUS LANES ON TRAVEL TIME IN HETEROGENEOUS TRAFFIC CONDITION USING VISSIM

S. Junaed[1*], N. Sakib[2], M.S. Mamun[3]

[1]*Department of Civil Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh, email: siamjunaed01.ce@gmail.com*
[2]*Department of Civil Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh, email: nazmussakib.austce@gmail.com*
[3]*Professor, Department of Civil Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh, email: mamun.ce@aust.edu*
[*]*Corresponding Author*

Abstract

The traffic situation in Dhaka has deteriorated to the point that it has become intolerable. For a large number of people, traffic congestion consumes a significant amount of time. It is vital that a large number of people can travel to their destination in the shortest time. Making public transit (PT) more reliable and efficient is one of the solutions to this problem. Introducing dedicated lanes to prioritize PT can be an option to reduce the travel time of PT, allowing a large number of passengers to be traveled in a short period. However, deducting a lane will impair the travel time of other vehicles. The purpose of the study is to evaluate the impact of this modification on the travel time delay of vehicles on the network. For this purpose, a roadway corridor in Dhaka city is selected as the study area where a dedicated lane for PT is implemented without increasing the number of lanes. VISSIM, a microscopic simulation software, is used for the simulation and the effect of dedicated bus lanes on the change of travel time of vehicles is observed. The evaluation is done in terms of the change in overall vehicle travel time, passenger travel time, and average travel speed of vehicles at the selected corridor. It is observed that implementing a dedicated bus lane reduces the travel time of PT by 37.6 % while increasing travel time for others.

Keywords

Traffic Congestion; Public Transit; Dedicated Bus Lane; Travel Time; VISSIM;

1. Introduction

Traffic congestion issues have received a lot of attention from many researchers as modern cities have developed[1-4]. The growth of urban areas is accompanied by changes in transportation modes and traffic congestion [5]. Modern cities are experiencing challenges in promoting sustainable transportation options to reduce the excessive use of private cars in urban areas [6]. In this case, Public Transit (PT) is the most effective way to move a large number of people, particularly in densely populated urban areas [7]. In reality, the functionality and attractiveness of PT are substantially affected by urban traffic congestion. Cost-effective solutions that do not require additional infrastructure are most desirable, and Dedicated Bus Lane (DBL) is one of them. But the prioritization becomes more difficult when it needs to be done in a road network with a poor lane-based heterogonous (mixed) traffic condition.

The traffic in Dhaka, the capital of Bangladesh is highly heterogeneous comprising vehicles of wide-ranging static and dynamic characteristics. The bus is a vital part of PT in the city of Dhaka. Homogeneous traffic has rigorous lane discipline and traffic entity types with little variation in their physical dimensions. In contrast, vehicles in heterogeneous traffic have a variety of static and dynamic characteristics, including size, engine power, acceleration and deceleration, maneuverability, etc. Because of the widely varying physical dimensions and speeds, the vehicles do not maintain lane discipline and instead occupy any available lateral positions. Furthermore, when different vehicle types coexist on the same road without any physical separation, the degree of vehicular interactions varies significantly with changes in the composition of the traffic [8].

The buses, being relatively larger vehicles, encounter difficulties to maneuver through the traffic and are frequently subject to acceleration and deceleration, which lowers the speed and causes discomfort to the driver and passengers. This also causes significant delays and uncertainty for bus passengers, which significantly lowers the level of service of buses and makes them a less desirable mode of transportation. In this scenario, in the absence of efficient bus transit, the potential bus users tend to use private cars and other paratransit vehicles.

As a result, to meet the travel demand the number of private cars and paratransit vehicles is increasing day by day which is causing traffic condition worsens as the space on the roads are limited. The implementation of DBL can be one of the most useful strategies here [9]. DBL allows PTs such as buses to move through a separate lane that allows the movement of buses to be unaffected by the change in the volume of other types of vehicles.

The objectives of this study are as follows:

- To evaluate the impact of DBL at the selected corridor in terms of vehicle travel time, passenger travel time and vehicle travel speed.

2. Methodology

This paper evaluates the effect of dedicated bus lanes on Dhaka city's road network using VISSIM simulation software. A curbside lane of the study area network was dedicated only to buses to improve the functionality of public transport. The study road network was modeled in VISSIM and after calibrating and validating the model, necessary data was collected from the software to evaluate the effect of DBL. Later the impact of DBL on vehicle speed, travel time, and passenger travel time was assessed.

Data Collection

This study was carried out on a major roadway corridor, Shaheed Tajuddin Ahmed Avenue, connecting the city's northern part with the commercial and economic center, Motijheel. This corridor is used daily by a large number of commuters to reach their destinations. The majority of those who use this route rely on public transportation. More than 7 bus operators currently offer their services along this road corridor. In addition, one of the city's main bus terminals, Mohakhali bus terminal, is located here, therefore in addition to intercity buses, long-distance buses also use this corridor. More than 70% of the vehicles in this corridor are cars and motorcycles, which negatively affects the movement of buses since lane discipline is not maintained (Figure 2). Currently, this road network is a two-way road consisting of 3 lanes in each direction without any priority for public transit. Data was collected from two different locations on the study road network on two consecutive days (Figure 1).

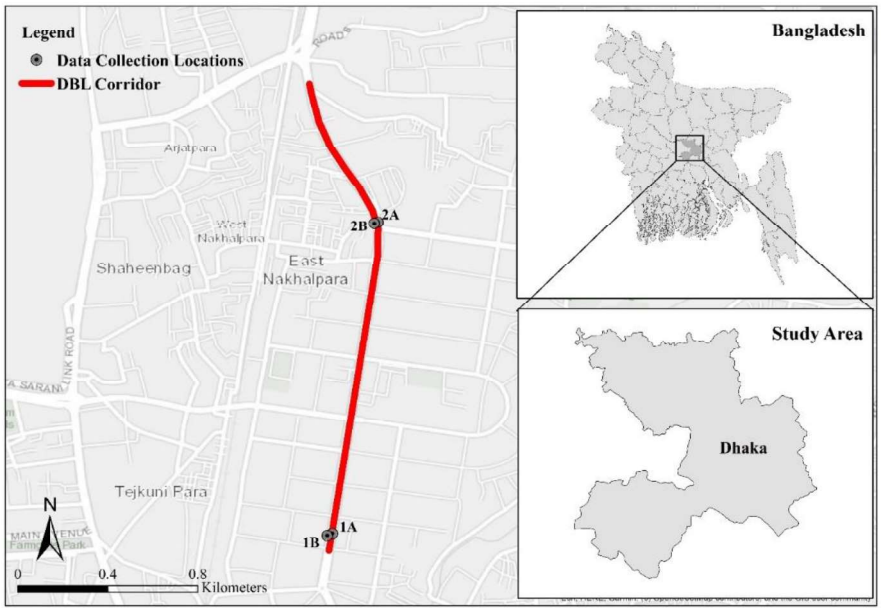

Figure 1: Study area with selected corridor for DBL

During the survey, a mixed composition of vehicles was observed. The vehicle composition of the study road network is illustrated in the pie chart (Figure: 2). Also, the passenger capacity of all types of vehicles on the road was reckoned. Table 1 presents the passenger capacity of each type of vehicle. The volume of vehicles from the north to south direction was observed to be 2489 veh/hr and 2821 veh/hr in the south to north direction.

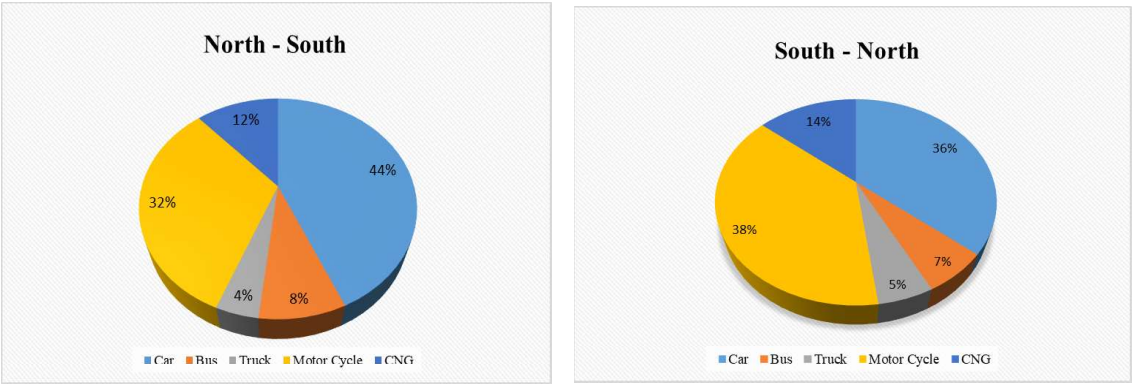

Figure 2: Vehicle composition of the study road network

Table 1: Passenger capacity of different vehicle types

Vehicle Type	Passenger Capacity (Nos)
Bus	45
Car	3
Truck	1
Motor Cycle	1
CNG	2

Microsimulation Modelling

A microscopic model of the study road network was modeled in PTV VISSIM 2021, a microsimulation software. The traffic condition of the road network is heterogeneous, oversaturated, and does have a mixed vehicle composition. In such traffic conditions, VISSIM performs better than other microsimulation software [10]. Default driving behavior parameters of VISSIM create a homogeneous lane-based traffic condition that does not represent the traffic condition of the study network of this research. So, the actual traffic condition was developed in the model by calibrating it by changing different driving behavior parameters in the software. Geoffrey E. Haven (GEH) statistics and a visual audit of the model were used to calibrate and validate the model [11].

The GEH Index is calculated as follows:

$$\text{GEH} = \sqrt{\frac{2(M-C)^2}{(M+C)}} \qquad (1)$$

In this study, the accepted value of GEH was considered less or equal to 5 [12]. Table 2 presents the comparison of field traffic volume with the observed volume in VISSIM during calibration and validation of the model. Here, on all approaches, the GEH value was less than 5 which represents the model's efficiency.

Table 2: Results for Calibration and Validation

Location	Calibration			Validation		
	Field Volume	Model Volume	GEH value	Field Volume	Model Volume	GEH value
Mohakhali	2489	2406	2	1693	1795	2
Satrasta	2821	2670	3	2426	2403	0
Bijoy Sarani	1145	1045	3	985	937	2
Love Road	220	230	1	190	205	1
Niketon	500	509	0	430	432	0

Dedicated Bus Lane (DBL)

In this study priority in space was provided for public transport in the study road network by providing a dedicated bus lane. Dedicated bus lanes are bus-only lanes that are set apart from other traffic. Dedicated bus lanes are placed on roadways (along the median, offset, or in a physically separate lane) that allow buses to operate across streets without being impeded by other vehicles [13]. In this study, the curbside lane was selected as a dedicated bus lane to ensure the convenience of the passengers. Additionally, there is not enough space along the chosen corridor for bus stops to be placed along the median. Figure 2 illustrates a schematic layout of the road in the study area with an exclusive bus lane.

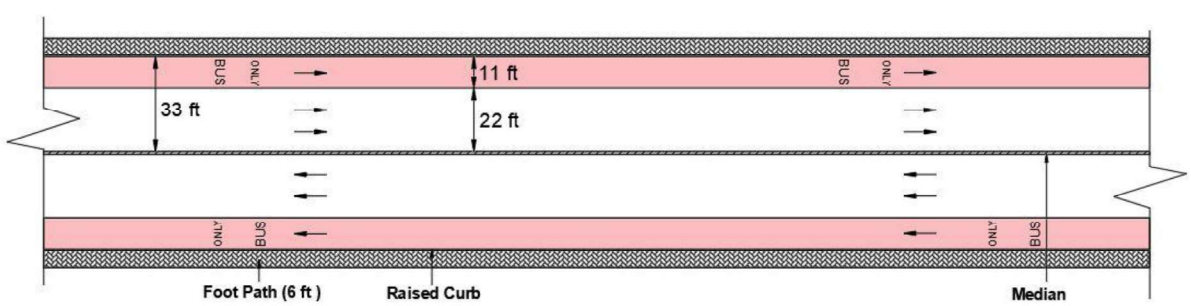

Figure 2: Schematic layout of the road with an exclusive bus lane

After validation of the model, the speed of all types of vehicles, travel time in terms of vehicle, and travel time in terms of passengers was measured and compared before and after implementing a dedicated bus lane on the road network.

3. Result and Discussion

The travel time required for the vehicles to move from Mohakhali foot-over bridge (M) to Dhaka Polytechnic Institute (DPI) in both directions was measured from the validated model with and without a dedicated bus lane. Then a comparison was made to observe the effect of DBL. Table 3 shows the travel time required by different types of vehicles to move from M to DPI in both directions and the change in travel time due to DBL. Here it can be seen that the implementation of DBL decreases the travel time of buses, however the travel time of other vehicles increases. Table 3 shows the change in travel time of the vehicles with and without DBL. Overall, the travel time of the buses was reduced by 37.6% in the selected corridor. Furthermore, for other vehicle types, the travel time increased by 20.18%. Table 4 presents the change in space mean speed of all types of vehicles in the network due to DBL.

Table 3: Travel time with and without DBL

Vehicle	Travel Time, Sec without DBL		Travel Time, Sec with DBL		Change in Travel Time, Sec		Change in Travel Time (%)	
	M-DPI	DPI-M	M-DPI	DPI-M	M-DPI	DPI-M	M-DPI	DPI-M
Bus	550	356	285	260	265	96	48.2	27.0
Car	524	336	687	393	-163	-57	-31.1	-16.9
Truck	575	355	662	427	-86	-73	-15.0	-20.5
MC	512	339	644	397	-132	-58	-25.8	-17.0
CNG	568	376	698	423	-130	-46	-22.8	-12.3

Table 4: Travel speed of vehicles on the network

Vehicle Type	Vehicle Speed, kmph without DBL		Vehicle Speed, kmph with DBL	
	M-DPI	DPI-M	M-DPI	DPI-M
Bus	13.4	20.7	25.9	28.4
Car	14.1	22.0	10.8	18.8
Truck	12.8	20.8	11.2	17.3
MC	14.4	21.8	11.5	18.6
CNG	13.0	19.6	10.6	17.5

From Table 5 it can be seen that approximately 64.16 percent of total passengers use bus as their mode of transportation and only 35.84 percent of passengers use other modes of transport. As the travel time of buses decreased significantly compare to other vehicles travel time increase, So, in terms of overall passenger's travel time, DBL effectively improves the condition of the corridor.

Table 5: Change in passenger travel time due to DBL

Direction	Vehicle Type	Volume	Passenger Capacity	Total Passenger	% of Passenger	Change in Vehicle Travel Time (%)
North - South	Bus	209	45	9405	66.52	48.2
	Car	1081	3	3243	22.94	-31.1
	Truck	104	1	104	0.74	-15
	Motor Cycle	804	1	804	5.69	-25.8
	CNG	291	2	582	4.12	-22.8
South- North	Bus	182	45	8190	61.8	27
	Car	1017	3	3051	23.02	-16.9
	Truck	152	1	152	1.15	-20.5
	Motor Cycle	1080	1	1080	8.15	-17
	CNG	390	2	780	5.89	-12.3

4. Conclusion & Recommendations

The VISSIM software model was implemented as close as possible to the actual conditions as all geometric and traffic characteristics of the area were considered when creating the model. Model calibration and validation was carried out to make it suitable for the particular purpose of this study. Hereafter, DBL was implemented in the validated model in VISSIM and its impact on the corridor was observed.

Overall, the implementation of Dedicated Bus Lane (DBL) on the corridor was positive. DBL improved traffic condition by reducing overall passenger travel time in the corridor. DBL ensures uninterrupted flow of buses, and improved its travel time significantly. Even though other vehicles travel time increases but it was also observed that most of the passengers at the selected corridor use bus as their mode of transportation. As a result, the overall travel time of passengers at the corridor reduced. Additionally, by implementing DBL, passenger service quality of bus can be improved which will help to attract even more passengers to choose bus as their mode of transportation.

Since the study focuses on DBL only, further studies can be performed by providing transit signal priority along with DBL as well as different variations of DBL such as intermittent bus lane.

Acknowledgments

The authors would like to thank the PTV group for providing the academic version of PTV VISSIM for the research.

References

1. Nagel, K. and M. Schreckenberg, *A cellular automaton model for freeway traffic.* Journal de physique I, 1992. **2**(12): p. 2221-2229.
2. Chowdhury, D., D.E. Wolf, and M. Schreckenberg, *Particle hopping models for two-lane traffic with two kinds of vehicles: Effects of lane-changing rules.* Physica A: Statistical Mechanics and its Applications, 1997. **235**(3-4): p. 417-439.
3. Zhu, H., L. Lei, and S. Dai, *Two-lane traffic simulations with a blockage induced by an accident car.* Physica A: Statistical Mechanics and its Applications, 2009. **388**(14): p. 2903-2910.
4. Zhu, H., *Numerical study of urban traffic flow with dedicated bus lane and intermittent bus lane.* Physica A: Statistical Mechanics and its Applications, 2010. **389**(16): p. 3134-3139.
5. Lu, J., et al., *Expansion of city scale, traffic modes, traffic congestion, and air pollution.* Cities, 2021. **108**: p. 102974.
6. Dell'Olio, L., A. Ibeas, and P. Cecín, *Modelling user perception of bus transit quality.* Transport Policy, 2010. **17**(6): p. 388-397.
7. Rahman, F., et al., *Perceived service quality of paratransit in developing countries: A structural equation approach.* Transportation Research Part A: Policy and Practice, 2016. **93**: p. 23-38.
8. Arasan, V.T. and P. Vedagiri, *Study of the impact of exclusive bus lane under highly heterogeneous traffic condition.* Public transport, 2010. **2**(1): p. 135-155.
9. KIM, H.J., *Performance of bus lanes in seoul: Some impacts and suggestions.* IATSS research, 2003. **27**(2): p. 36-45.
10. Mashrur, S., et al., *Modeling Modified Intermittent Bus Lane Integrated with Transit Signal Priority Under Mixed Traffic Condition.* Transportation in Developing Economies, 2022. **8**(1): p. 1-18.
11. Agency), U.U.H., *Design Manual for Roads and Bridges. 12, Section 2*. 1975.
12. Mitkas, D.-Z. and I. Politis, *Evaluation of alternative Ramp Metering scenarios on freeway on-ramp with the use of microscopic simulation software Vissim.* Transportation Research Procedia, 2020. **45**: p. 483-490.
13. www.Weebly.com. *Dedicated Bus Lane(DBL)*. [cited 2022; Available from: https://planningsustainable.weebly.com/dedicated-bus-lanes.html#:~:text=A%20dedicated%20bus%20lane%20is%20a%20lane%20for,move%20through%20streets%20unobstructed%20by%20other%20vehicular%20traffic.